

# ActInf OrgStream #001.1 ~ Richard D. Bartlett

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introducing myself i can go on at great length i'll try to be brief um i'm from new zealand that's why my english is weird i currently live in italy though um because i wanted to be connected to everything that's happening in europe um i studied engineering at university so i've got that frame of reference of thinking in terms of systems and modular parts and how things plug together and in the last 10 years i kind of drifted away from technology and that kind of innovation and got more interested in what you might call social technology so how do we organize people what kind of systems support them to work together effectively but with a different set of values that are usually applied in most organizations so instead of for instance prioritizing efficiency or quality or you know how quickly can we deliver stuff i'm interested in some other values which are more like how do people develop as humans how do we create relationships that really make a difference in our lives you know how do we pursue something that feels meaningful and has a positive impact those kind of values and then and then what kind of organizational structures support those kind of outcomes so that's my jam i've done that in lots of different formats so building a startup company working in an entrepreneurial ecosystem which is all self-managed with a couple hundred people and now i do a lot of um i guess designing general purpose community and organizational structures that can be reproduced in lots of different contexts thanks for the intro and i guess here on the slide we have one of your ongoing projects patterns for decentralized organizing just everywhere in there it's pretty cool it's like pattern languages and decentralization and organization all topics that are important inside of active inference as well as outside and in this work as well as in a episode 51 on the jim rutt show you talked about this multi-scale view on social systems so we're going to kind of structure this conversation along this ladder of nested systems or nested processes we might call them nested markov blankets in active inference so maybe could you go over this multi-scale nesting of social structures from the individual on up and why is it important to think about it that way sure so this is a lens that you can put on to understand social physics and it's not um not making any claims about defining reality or something like that but it is a really useful lens to put on to like what is going on in these group dynamics in

this organization in this community so the smaller scale of the individual and even at that scale the one person it's useful sometimes to think of a person as a group to think about i'm not just richard i am all of these different tendencies all these different personalities affordances parts there's different parts of me some some of my parts are very enthusiastic and confident and bold other parts are anxious and withdrawn you know there's like all these different tendencies going on and it's interesting and useful to think about the metaphor between like there's a kind of parallel between the way that my parts are related to each other and the way that i relate to other people so there's like a kind of internal relationship going on which is a metaphor for the relationship that i have with you so if i am very um you know constricted and defensive around if i want to disown my anxious parts if i want to pretend like i don't have anxiety and just kind of like remove that whole part if you show up as my friend with anxiety i'm going to be defensive against that too it's like if i can't accept my anxiety i'm not going to be able to accept yours so this is a um a metaphor that's been really fruitful for me i i kind of came across it through some amateur theorists and friends of mine just thinking about this way and then discovered that actually it's used in um therapeutic context so there's like transactional analysis internal family systems probably a few other systems i don't know so well therapeutic modalities which which kind of treat the individual as a group and use that as a as a metaphor for getting insights into what's going on in a person's life so that's one that's one dimension and and as i described that it sort of implies the way of working that i'm interested in this more decentralized participatory collaborative shared power structures etc it seems to involve or require interior curiosity that there's that that the all of the agents have to be curious about themselves and the impact that they're having and what are their motivations and values and drives and why did i get really upset when that happened and those kind of questions sort of have to be on the table so then the next scale is the dyad and a dyad is just two people at this point i think it's really useful to pull in rihanna eisler who's this amazing theorist author historian um and she has she's she's named this framework which i find really helpful which is the partnership nomination spectrum and it's a way of saying like oversimplifying obviously all of history and all of human relationships and saying there's a spectrum in any relationship which can be more like a domination or more like a partnership so more partnership being we're different but we're equal in some important way we've got different competencies that we bring together and a combination saying there's like a ranking system where one person has more power than the other one person is kind of controlling the other the other person is submitting and in any interpersonal relationship we can be anywhere on that spectrum from

very powerish to very domineering and not just at the interpersonal scale but also at the very large scale so she she in a lot of her work she's described the connection between for example what's happening in nazi germany in the 30s and 40s and what's happening in german families at that time you know how did those how do those two scales interact with each other um so the diet let's let's assume you agree with my my values which is i'd like to live in a culture that has less domination less submission more partnership more equity um the diet is the first place really where we get to practice that to say when i'm in a relationship with you am i trying to assert dominance or can i find a different way to relate can i you know do we always have to compete and make a ranking or can we find different ways of linking and making this like multi-dimensional connections which are not so focused on who's the best and the strongest and the coolest and the fastest and the smartest uh but just let's discover all of our unique competences and see how they fit together in interesting ways so that's a diet scale and if you think about a really great friend or a coach or a therapist or a partner those are really crucial dyads if you're interested in growing and developing and maturing or evolving i think a lot of that developmental work happens in a dyadic relationship i think there's often someone else there who's like you know sometimes we say holding the space you know it's a very vague term but they're they're creating some kind of nurturing container for you to do your thing then the next scale up is called the crew and i use the word crew because i was thinking about a sailing ship you know it's a very active people have different roles there's there's this expectation that we're different but we're also doing something together it's practical it's hands-on and a crew is small i usually say it's the size of a dinner table conversation so like five people maybe and i think there's actually some good science on this now i was listening to a recent podcast with um robin dunbar and he says that five people can have one conversation because they can all model each other's perspective simultaneously and kind of hold a shared state and and it's kind of we can our brains can kind of natively manage that but six or seven people they just can't hold everyone all the different you know the combinations of that gets too too much complexity and so you'll notice this if you're having a conversation with five people and then two others show up that the conversation will split into multiple sub-conversations just completely natural happens all the time so the point of focusing on the crew scale is saying there's something that we can do with a group of four or five people that we can't do with eleven we can't do with seventeen we can't do with two you know there's unique affordances and competencies for that group for that size and and a big part of what i'm doing with this framework that we call micro solidarity is trying to train people in a literacy of scales

and understand there's an articulation at each of these different scales the groups are good for different things you know um if if you want to have an intimate conversation probably a couple of people but if you want to have like a dance party you're going to need more you know it's like we can do different stuff with different groups and the crew is like when people are starting out in micro solidarity practice the crew will often be a space for peer support belonging connection um you know maybe you take a course together and they're your buddies that you're thinking through this new information it's it's kind of about connecting and then once those connections are established then it might evolve to the next stage which is much more about action getting stuff done um having some kind of shared investment collaborating on projects that really mean a lot to you starting companies uh that's that sort of activity and so then the question is how do you find your crewmates because you don't have chemistry with everyone we don't we don't all fit together with everyone and that's what what i call a congregation that's what that's for so the congregation is um could be 30 people but i kept it around 150 or 200 people so a group that's big enough to have some diversity um but small enough that there's no strangers there that everyone has some sense of who everyone else is they might not be super close but we're on first name basis with everyone and it's called a congregation because you congregate that's the main thing that happens is these um occasional gatherings where people come together and you get to meet all these other people and it's like a dating pool where you can find your crewmates and this could be a conference it could be a retreat it could be you know a summer holiday there's lots of different ways of doing this but again there's the scale factor where we say there's a boundary here around 150 people maximum because again there's some things you can do with a group of that scale that you can't do with a group of 500 and especially if you're doing governance you can govern a group of 150 people basically with dialogue trust mutual exchange of like who is this person what do they care about like i can we can have good enough rough models of each other and then maybe articulate a few agreements or a few principles and that's kind of enough to keep the system coherent and organized you don't need a whole complex bureaucracy you don't need a lot of checks and balances you don't need some kind of legislation system or a police service or any of that sort of stuff which you do as you get into the larger scale so it's saying let's draw a boundary and say there's a kind of efficiency that happens at that scale and then in my framework at least for now anything bigger than 150 people is what i call a crowd because once you get beyond that scale you you can just assume that most people are going to be strangers to each other so you might for example go to a festival or a conference with 5 000

people and you might feel some sense of imagined community some sense of like these are my people and that can actually be a little bit confusing because it feels like they're your community you don't actually have enough mutual knowledge of each other that you can do stuff you know so like if you if you don't show up one day because you're sick or there's some dramatic thing happening in your family you can't really count on those people to look out for you because there's there's just a lack of visibility you know there's just too many too many relationships to track and so i just put everything from 150 up to 8 million in the box called crowd and it's kind of like that's a that's a problem for someone else to solve when i'm focused on the small scale thanks for the overview and there's many multi-scale systems and active inference ideas that we'll return to and i think how those different levels of organizations play out in the past present and future of science and research and in other domains i think we'll return to that but let's go to relationship first so here just to sort of bring the active inference into play is a figure from our uh 2020 paper on active inference and online teams and online organization so active inference as the name suggests is about action and inference it's about how perception in the context of a generative model of the individual translates into action and here we have two teammates so this is really getting at that minimal relationship the minimal dyad and each teammate is perceiving the digital information that's their shared informational niche and then they're each perceiving it uniquely unpacking it in light of their own experiences and then acting and that's modifying the niche so this is like two people on a video chat or editing a shared document and this is sort of an overview that is reminiscent of many other perception action loops although in active inference there's a lot more math and technical details that underlie it how do we design for healthy interactions and multi-scale resilience especially online knowing that there's so many degrees of freedom and so many ways to just each little it point and interface in this diagram how do we know how to design what works this is so great i love looking at this diagram and just like doing the translation between the cybernetics and and what i'm into um i mean what first comes to mind looking at this diagram is the two the two speech bubbles as they come closer together like in common parlance we call that intimacy i think like when we when we get closer and closer in our shared context where i can trust that i know you and you know me and that we actually do have shared understanding that um when i'm saying something you've got more or less the same picture in mind as what i've got that's intimacy and that's um this very i mean the idea of like how do you engineer intimacy is a very clumsy concept you know it's very unfamiliar concept but that's a lot of what we're doing i think another thing that comes to mind looking at this diagram is

you've got the thought bubble of perception of course the perception includes the other person and it also includes your imagination of what the other person is thinking about you and and i mentioned interior curiosity right this is all part of the whole system of when people are starting out in this way of working in this participatory collaborative shared way of working one of the things that often needs to be developed is self-awareness meaning do i know what impact i'm having on the group so it's not just me broadcasting like i've got these things to say and i just show up it's also the kind of tracking what's my impact how do other people perceive me um and and does my self-perception match their perception and and there's always a gap there what do i do about that gap and how do i bring those two things uh closer into alignment and again in my experience that happens in relationship that's not something you can really do on your own maybe some people know how to do it i certainly don't know how to cultivate a great deal of self-awareness on my own it's always in these feedback groups with other people that are that are helping me to tune in my self-evaluation there's some very concrete practical stuff you can do to improve self-awareness and relationships and that can be like um sometimes it almost sounds maybe naive or childish or something but often we'll start with the very concrete what we call listen and loop so i'll share something and you'll reflect back to me what you heard as close as you can track it and then maybe the next loop would be i'll share something and then you'll share what you heard was important to me some kind of interpretation or some kind of abstraction of what you heard and we're and we're really as i hear what you're saying back to me um i realized ah that is what i said but it's not what i meant and i need to add this extra information and there's this tuning process that's going on between us so we do a lot of that a lot of really intentional like here's a process do this then do this then you know you're gonna have five minutes of this person sharing and then three minutes of this person repeating back and we kind of drive people through these very um defined choreographies and that very quickly helps people to see oh when i'm communicating people are not tracking what i'm saying you know the thing i'm the reflection i'm getting back doesn't actually feel very close to the picture i had in my head and it helps you tune that and basically learn how to communicate more effectively in active inference we talk a lot about having a shared generative model and there's another figure where narrative is playing the role of a shared generative model and that concept of thinking through other minds which is like our cognition is intrinsically about others world models as well as it including their model of us so these are all um important topics and things that again we kind of have the space at the high level mapped out the possibilities are there in the formalism and then which islands of

parameter combinations actually lead to the kind of positive social relationships that we want to foster that's the empirical question and then also to connect that to the so the way that policy selection or basically behavioral selection works in active inference is it's always a combination of the pragmatic and the epistemic value so like what is the utility what is the external motivation of this action and then what information do i gain what is the knowledge gain um or the intrinsic motivation of the action and then just like you said with a choreography it's like contact improv that's at the physical level there's sequences of actions that dyads can participate in over the hour over the month over their lifetime so that they can learn how to have a better joint model of inference and action so that they can do the tango better or so that they can do some sort of physical operation better choreography that we see in the operating theater and then we take it into the realm of symbolic communication and so it's easy to see when your hands don't match or you look at the video you look at the the golf swing but then in communication it's a little bit of a less tangible space and so it's really important to have these kinds of practices that help us calibrate our meaning transmission i want to add on this you're getting my brain going on so i just talked about the diet scale but obviously i'm always passing it through other scales and seeing how do we play there so at the crew scale um what we discover is that in any group you will have a unique contribution to make to that group you'll be the best at something you'll have you'll be the most ready for some something and you don't know what that is people don't know what that is they discover it through the reflections of others so that's that's um [Music] that's a beautiful process actually but in the larger scale when you get up to sort of 30 people or 50 people uh there's something incredible that happened to me like my whole process in this accelerated hugely through my involvement in the occupy wall street movement where we were doing a lot of consensus decision making with 30 to 50 people sitting in circles day after day after day of the day and what was happening there was um you know we had a camp and we were trying to organize our camp and people needed a place to stay and to eat food and then the media would show up and have to talk to the media like there's all these decisions that need to be made as a community and we decided we're going to do it by consensus which means someone brings a problem or a proposal or a need and we're literally sitting around in a circle and it's as if as you hear from all the different people you're just adding more and more perspectives like these little glimpses into truth that i can't see on my own but with the right kind of group process and listening and like there's a ways of assembling all of that knowledge into like a three-dimensional picture and actually it's like a 50-dimensional picture when you've got 50

different people with their own different models and experiences and and i got so motivated by that because it was obvious that the collective intelligence when the conditions are right the collective intelligence is so much superior to any individual intelligence like you the group with the right process the group is always going to be the smartest person in the room and that is so motivating because once you see the the the brilliance of the intelligence and the creativity and the you know the joy of it it's there's so much enthusiasm that gets kicked off it's like oh i want to be part of this thing and now me as an individual coming back to the individual scale i often get feedback from people saying that like in my blog post for example that i that people value the way that i articulate complex things in simple ways the way that i can put into language experiences they've had but they've never quite known how to articulate and the reason i can do that is because of all of that consensus practice is because i've brought proposals and had them disagreed with and challenged from all these different perspectives and learned through many iterative loops how do i communicate in a way that many different people with all of their different ideological baggage are going to be able to listen and recognize the thing that i'm talking about um and that's now one of my favorite skills that that i get to bring to the group is like oh i know how to articulate stuff without alienating people along the way and so it's like we're going up and down the scale all the time with this developmental process awesome so you mentioned the person as a and the the organism is involved in unified action so if you know one leg's walking backwards one's walking forward just gonna rip in half so we're involved in unified action that's when the individual is acting as an individual and consensus governance is a protocol for generating unified not necessarily unified inference because you can have consensus it's like yes i consent that our co-op will buy almond milk i have a different take on that than somebody who needs to buy it for a health reason but that's okay we have a different unpacking we have a different perception but we're involved in shared and so that's a very key point that consensus allows us to have groups act as individuals but not necessarily in thought indeed and that that may not scale to all levels if one veto can take down a uh a whole congregation's actions that's a very different question so how do governance mechanisms change across scales is definitely something you've um brought up let's go to this do you want to add anything or we can go to yeah one thing on this on this organism like this is really animating me at the moment um so in your body it's clear like if i need to mute the microphone because there's a noise going on i just need to move one finger right like that that's the the action here if i need to pick up my water bottle that kind of takes my whole arm it's going to do it's got to have this coordinated action and let's say i need then i



need to go to the bathroom well i'm going to have to move my entire body and it's going to have to move as a whole and this is where consensus kind of breaks down is because it usually treats the group as a as a single limb and what we need is the articulation to be like there's some things that are like go to the bathroom where you need to move the whole group but there's lots of things that are more like switching off the microphone or taking a drink where a much smaller subgroup is the appropriate site for the decision making to happen and it's really difficult if you're used to doing everything with the full group to then break off into small groups without it triggering a whole bunch of conflict and alarm bells and that's basically what i'm trying to do like the very biggest picture of what's going on in the world how do we pull through with climate change and inequality in society and all these things that are going on i think what we need is the ability to articulate different scales of coordination something like climate change we need the whole global body to move and there's some things that are happening in my neighborhood where we just need one we just need the wrist and the fingers you know and we need to have in order to be able to do that we need to be able to address these different groups at different scales and and that means we need to have like a nested collective identity that goes all the way from me and my family and my flat and my building and my street you know that kind of way wear them in address uh it needs to go all the way up to the global scale i need to be able to sort of identify who am i globally who am i in the in this bio region and and working at the small scale at least let's track the self the crew and the congregation there's three scales at least can we get the competency and the literacy there and maybe that will equip us to be more familiar and coherent and literate at the larger scales that's kind of the big project okay very cool and just thinking about like a active inference lab or the field we have individuals who have buddies in the lab we have crews that work on specific projects the organizational unit is more verging towards the congregation there's people who are on non-overlapping projects and then a field is like a crowd people check in and leave to scientific fields all the time and no one knows or cares all the time so it's like a music festival like you brought up earlier so let's continue our walk up the ladder although i suspect that we're going to close the loop um there's a few topics here and uh dean um and i spoke yesterday so it was a good convo and we got a lot of these things down so first let's think about this crew and congregation levels this idea of a threshold model for collective action and the way that's related to skin in the game so previously i studied foraging regulation in ants and we used threshold models for ant decision making and they quite literally have exoskeleton in the game when they forage as nest mates and the colony has foragers on the

line when they send them out and then it's like they take the model from neuroscience with neural activation and we apply it to the behavior of a different system and then here we're thinking about threshold behavior in human systems and collective action across scales so where does threshold modeling come into play for group decision making in people so good no one's asked me that before i hope i understand the question well enough i'll try um one thing that was really striking to me in the first so i'm in italy which is the first kind of western country to have to deal with the covert pandemic and then there was kind of maybe a month you know times dilated but i think it was a month or two where covert was very active in europe but i hadn't arrived in the states yet and then there was this moment as it was kind of people were realizing the seriousness of it people in the states weren't dealing with it yet but they were awake to the threat there was this this couple of weeks a window of opportunity where i'm quite active on twitter i do a lot of connecting with people there and thinking out loud there and it felt like the entire network of twitter was all focused on the one topic of there's a pandemic coming how do we get prepared what's the truth what do we know what do we don't know all of this and because everyone had skin in the game it was such a good conversation that i was connecting into it was such high quality decision making and feedback and hey we discovered this and this is a model that you can use and did you realize that this whole thing is corrupt and all of this was happening very very rapidly and very efficiently because everyone was really they had skin in the game because they were thinking about my family is it is at risk here how do i keep them safe it was very meaningful to find this connection between the global pandemic and the very local immediate the choices that i make may be the difference between life and death for the people i care about and that was the the most uh coherent and high quality exchange i've ever seen on twitter at that scale now most people don't experience twitter that way most of the time usually people experience twitter as a much more polarized um you know there's a lot of in-groups and art groups and kind of battles around these political ideologies and culture walls and so on and i think a lot of that is about this lack of skin in the game to a degree all this confusion about skin in the game there's a there's a mismatch between my sphere of influence and my sphere of awareness so the last few decades of information technology have expanded my sphere of awareness to a global level where i can be aware of this horrible atrocity that just happened in this small country that i'd never heard of until yesterday um but my sphere of influence while it has grown like we've been we've got all of these extra capabilities because the internet it hasn't grown nearly as far as my sphere of awareness so it's like i'm continuously absorbed by all of the stuff that's

happening and i've got all of this stimulation coming in but my impact is still relatively small it's much smaller than global i can't do anything about that atrocity that's happening in that small country i've never heard of before even though i can feel a lot of stress and like it's really important to me and i've got something to say on the topic and so that is part of what i think we're seeing happen on twitter is if someone is not really clear on their own agency like what impact can i have what decisions are up to me where's my autonomy how do i how do i achieve things in the world how do i do action that's effective not just as an individual but with a committed crew of people that are doing it with me or with a neighborhood or a community of people that are doing it with me if you don't have that experience of agency and autonomy at those smaller scales it's very easy to just kind of get distracted by the global scale and have an opinion about everything and make everything into a big drama and debate and organize yourself into these polarized camps and say i'm on the good team and everyone else is a bad team it's really easy to just kind of slip into that and you're not achieving anything a lot of the time you know it's like this feeling of participation without actually any impact and once you get activated once you discover your own agency once you discover a few committed citizens who are going to change the world with you once you have a company or a crew or whatever then suddenly you realize okay this is what i can achieve like this is what's realistic i can change the consumption habits of my neighborhood that's something i can do i can't change the whole global economy that's not really within my sphere of influence and so i'm going to spend more of my time focusing my intelligence and my energy on the stuff that i can make a difference on and and that's another part of the project here is to get people activated into their agency and once they do that they focus on the stuff where there's skin in the game and and have a bit more i don't know if it's like it's like humility or modesty or something or a little bit more of like i don't actually know what to do with the global scale i'm going to relax a little bit on my intense opinions about how i know what's the right thing to do and just focus on where i can have a meaningful impact very cool so one key difference between active inference and a lot of reward centric frameworks is that in contrast to thinking that the maximization of reward is the imperative for action we frame the reduction of uncertainty as the imperative for action so we can update our models that's learning or we can act to reduce our uncertainty and that's what motivates behavior and it has that pragmatic component which is like utility a reward but it's not just simply reinforcement or reward learning so we can connect that to what you were saying there is so much uncertainty in the world there's things we just simply don't know about we don't know we don't know about and now that recently we have been able to have our

sphere of awareness expand to the global or even the intergalactic scale that breeds uncertainty of action and so there is that overlap between our sphere of influence and our sphere of and we can think about how can we reduce uncertainty if only to reduce our anxiety or feelings of negative valence but actually to be more effective as well locally in our local situation we can reduce our uncertainty about the um volatility of the housing conditions locally and then also um aligns us with just other people who we then actually in a bottom-up way start to be able to do more and more people start connecting bringing their perspectives and skills together the awareness and the capacity for action is increasing because you entered at the level of individual agency so that's like a very important step that when we activate we are also starting to build from the bottom up rather than just looking at how high the mountaintop could be or looking at how many mountain tops there are and there must be infinite ways to get there as well i want to i want to rip on this uncertainty thing that's really i've never heard of that it's a really cool way to think about it um so we talked about how the sphere of awareness is growing and we kind of have been using these special metaphors but it's not just that there's more coming in it's also that there's more incoherence so we used to have this thing called the six o'clock news and you would turn it on and it would say these are the things that happened today that are important for you to know about and everyone was referring back to the same six o'clock news and then they'd go to work in the next morning and then have opinions they might disagree but they were all referring to a shared baseline and the other thing that's happened with the internet is that we don't have six o'clock news we have all the time news and it's coming from many different perspectives and they all disagree with each other and so we've lost that shared baseline now i don't think that six o'clock news was ever telling the truth or was particularly representative of reality but it was simple it did provide at least here's our zero line and we can discuss here and there about it um so there's so much more complexity of perspective that we're having to confront as well and so this the sense of uncertainty is just magnified so much not just because the volume of information but the incoherence of it and so how do we create what's the shortcut to create certainty there's two that i know very well which you see online all the time one is moral superiority like me and my people are right there we go solved all of the problems the people that are like me are the ones who are right all the other ones are wrong they're ignorant they're stupid they're evil like that's bull efficient and then the other one is conspiracy theories where me and my little cabal of people who have somehow access to this superior knowledge have worked out there's a plot if you just connect all the dots you'll see how this is definitely what's happening here's the plan this is who's

pulling the strings this is why you've just got to follow the money man we've got you know like that's really flourishing at the moment as well and it's this desperation to cling on to a certainty and an incapacity for the degree of uncertainty that actual reality actually looks like when you start hearing from all the different people simultaneously so in active inference the preferences are over sensory states so that's what ties it to perceptual control theory like we want to perceive temperatures that we can live with it like homeostasis but to a large extent the inference is on unseen causes of the world and so there's interesting work about bayesian views on conspiracy theory where you can have a simple model that explains all of the observables so that's um a nice point and another way that we can sort of shortcut the uncertainty is we can make nested models like agree to disagree so rather than uncertainty at the sort of lateral level you can encapsulate that in a model that both people can actually have shared precision on which is like we agree to disagree or um you know this is a really um wildly inaccurate thermometer but we're gonna kind of bound that within a way of thinking about that thermometer so we're not gonna be stressed out when the thermometer goes up and down we have a precise action strategy for dealing with this thermometer so then we start to identify or seek precision at a higher level and understanding that there's volatility at a lower level you wrote a really cool 2019 medium post i will if you will there's almost a few readings of even the title because on one hand like i will do that if you will do that but also to will is to want so it's like i want to if you want to so that actually speaks to how we want need to have the shared action thresholding but then there's that thinking through other minds shared intention mapping how does scale or work in the dyad you know in the i do's that's the sort of minimal unit of i will if you will go to these levels like the crew the congregation maybe even the crowd to start thinking about i will if you will bringing skin in the game and having real effective action i think this is legitimately one of the the most promising places for communication technology to make a big difference on society and for the good listen to have a positive impact um and it's solving these kind of coordination challenges where one example that comes to mind i think actually from that article that kind of triggered this whole research is it's quite common that that tenants will have a bad relationship with their landlord and the the landlord has a lot of power and the tenants have very little power and so they'll often be in unfair conditions or they're being mistreated or it's you know it's somehow exploited of that situation but because of the way um the information is flowing in the way the relationships are structured there's there's kind of nothing that an individual tenant can do about it apart from finding another landlord often with the same problems but if the tenants could be coordinated

with each other then they can meet the they can kind of counterbalance that that power dynamic for example i will if you will if you could get everyone in a building or in a neighborhood they had a shared like slumlord landlord to say i will go on rent strike if you do like if everyone in the block agrees we're not going to pay rent until we see this improvement in our housing then you better believe that they're going to get the action that they desire if they can coordinate that agreement and then say to the landlord look everyone we've got this pledge here you can't evict 300 people like just fix the leaky roof you know um whereas if it's one individual that says to the landlord i'm not going to pay the rent well then you just get evicted and so i think that example applies to how do we reduce carbon emissions you know how do we um confront the really large challenges we face in society often there's a disadvantage to being the first mover or there's a disadvantage to doing something on your own without other people doing it with you and sometimes the disadvantage is so great that it's like it's meaningless i often say there's no such thing as a boycott of one you know i can i can decide to not fly but that doesn't have any impact on carbon emissions unless i'm participating in some kind of organized coordination movement where other people are also not flying with me and we're we're actively growing that shared power and we're saying look at all these people who've agreed not to fly we need to see electric airplanes please or more green forms of transit and to be able to counterbalance that force um and so that's the kind of thing that can very easily be solved with the right the right precise use of technology which is basically asking the right question of the right people and giving them the button to text that says yes if 10 000 people in my neighborhood agree to this action then certainly i would do it like i would love to do it but i'm not going to do it on my own and let's see if you can get that conditional commitment and if it reaches a certain threshold then yes we all act and that's the kind of problem that computers were basically designed to solve i think so that gives me a sense of a little sense of optimism yes um thinking about that flying example it's almost like there's an attractor and if you just pull out individually you've reduced the demand and actually that that might lower the price so now another person jumps in so as long as there's an attractor in the system then uncoordinated action is just part of the structure of that attractor how do we actually shift the nature of the attractor that's where the coordination comes into play and that's kind of what we wrote down with like web 3 and distributed autonomous organizations and different kinds of cryptographic technologies it's not just you're seeing 10 000 likes on something and that could just be basically a psyop on you from the side of a server how could that be verified information so you know those are real people who have made a commitment

to that's when it starts to get very interesting and then we wrote down a few active inference terms like thinking through other minds which we talked about the importance of the counterfactual because for any definite statements there's so many counterfactuals we can go off into and when people are even talking about what kinds of worlds they want to see it's sometimes a mixture of their preferences as well as counterfactual action policies it helps to have a formal model not the only possible way with active inference but a formal model so we can separate someone's preferences like what they do from counterfactuals on action other things we could do and then just separating the discussion that way helps us be really creative with the counterfactuals of action and ask which will actually reduce our uncertainty about realizing the preference so um that's hopefully going to be an active and interesting area with getting people structure commitment because without a lot of intentionality across different scales and careful design the internet is a low trust place so making it high trust at multiple scales is very crucial yeah the um the the piece about dao's web three i guess that has some promise i mean i should clarify i think the concept of a dao de centralized autonomous organization from my research over the last couple of months it means almost nothing like it doesn't really have a formal definition it's like a it's like a parenthesis that we put in the conversation which means all of my hopes and dreams about how organizations could be better which is a very useful object to have in a conversation but it doesn't tell you much about how to do it but i i'm very enthusiastic and optimistic about it because you've got these groups that are experimenting with resources that they really care about like a lot of the dows that are running have multi-million dollar treasuries that are being governed by tens up to thousands of people and they're confronting the real hard problems of how do you do that well one of the main bottlenecks is not like i described you know asking people the question and giving them the button to check it's the attention economy to actually get people coordinated to look at hey here's the button you need to click that's something that you want to spend that wisely and then and spiral in the community that i've i've spent most of my time in we've sometimes even talked about that the attention economy is kind of like a shared credit card and you want to be really responsible when you spend that you can't be putting up a vote every five minutes you've really got to be careful about that and that's something that um that's one example of a kind of bottleneck which requires a lot of sophistication and nuance and we don't really know how to program that yet that seems to be like a human strategic governance you know challenging communication problem totally agree about the sort of everything and nothingness of dao and i think maybe even the word dao is a subtle nod to that because distributed is on a continuum

and there's not a single definition autonomous it's also on a continuum like people will say oh the embryo is a self-organizing system oh yes but of course there's the system that supports it so nothing is really self-arising so that's a continuum and kind of has nuance and then organization which is what people and other systems have been doing forever so it's kind of just saying we're using digital technology to organize which is awesome and it's many of those discussions that we would have in the co-op living room again about the almond milk but now it's happening with digital scarcity and real assets and lives on the line and then you mentioned the attention economy so another active inference term is like regime of attention which is kind of just describing how attention and salience is being distributed through space and through time so just doing a mindfulness exercise in the body and okay pay attention to your feet and your knee etc that is a personal level regime of attention and you're speaking to collective regimes of attention and then where do governance decisions come into play are we going to micromanage are we going to try to get everyone's attention or might it be sort of a distributed regime of attention that doesn't require as much centralization good jumping off points for of course org number one in our last few minutes let's kind of close with the crowd and to uh paraphrase slash mutate a quote any sufficiently advanced crowd is indistinguishable from an individual so that's kind of where that multi-scale ladder loops back because of course we're an ensemble of cells and so on so this multi-scale nesting it ends up kind of closing a loop between a true crowd mob and then the individual because once it's a crowd it's just one you don't say crowds it's just a singular so that relates a lot to some technical discussions we have about mean field approximations like can you just assume that that cup has infinite water molecules just take the total average how do we contrast that with a unique trajectory based or a particle particle-based approach so those are physics questions but they come into play with this dialectic between individuals and crowds and then also what's outside of the crowd or what is the niche of the crowd so those were some crowd related thoughts we had any anything to say on that yeah definitely um probably the most inspiring examples trying to advance participatory democracy um in the decade that i've been paying attention to anyway they've been happening in taiwan since 2014. and one of the tools that they use is called polis which sometimes they describe as like a ai ml facilitator basically the way it works is you ask a question of a large crowd um or you you poll for opinions or responses people can basically say i'm close to this or i i i agree or i disagree that's basically the responses that they can do and then it clusters all these sentiments and says oh all of these people seem to more or less agree with each other and these ones agree with each other but there's a disagreement between



these two crowds and so essentially what happens is you take thousands of people and you'll see there's like three or six opinion groups it's like it you can boil the crowd very easily to a crew and and then what you need to do well first of all the crucial thing here is it's a rough consensus it's not consensus it's rough consensus it's like there's there's a cluster here you know like there's not a firm boundary but there's a there's a kind of coherence at and then once you've identified the four or five opinion groups you just need to find a trustworthy representative of each one bring them into a deliberation that's facilitated by a human and broadcast that conversation and and the five of them sense making together and uncovering okay what's our rough consensus between the five of us and you very rapidly generate very high quality policy that people can get behind to the extent that the quality of the policy development they're doing in taiwan those kind of processes will get rubber stamped by the appropriate ministry and and government to say yes we can we can improve this right away so that's like so inspiring to me and i guess the process of going from those five opinion groups to the one policy that everyone can more or less live with that's kind of the process of going from the crew to the individual you know it's like this is what the collective body can tolerate or this is this is what we have in common this is our island of agreement so that's like so incredibly exciting for me to to track that happening cool um i've heard of paulus and several other related tools in metagov and we talk about these different levels of governance that's the body politic and we can trust the process we can have high confidence and precision on a process to really that highest level where you just basically suggested that they have so much trust in it that whatever the outcome is whatever that thermometer reads we trust the thermometer and so we can have precision on that even if again so trusting the process way of kind of bounding our uncertainty i think this is a fun kind of closing notion for the first time that we're speaking here about theory and practice which is something that came up in multiple um podcasts and writings of yours theory and practice otherwise known as kind of inference and they're inseparable they're the two swirls in that tao how do theory and practice come together across these scales and just as we move forward how does our individual theorizing and practicing relate to these community levels um great question the the if you read the original micro solidarity proposal it's called a proposal because it starts with this is my plan of action this is what i'm doing that's that for me the verb always comes first and then oh along the way i'll have to explain some theory so you understand what i'm talking about but it's always for me it's really important to keep it grounded in action i think there's um i have a model in my head there's action and then there's talk about action and then there's talk about talk and i've got

very little capacity for talk about talk because occasionally you need to go there but i think it's very easy for a sort of culture to get once one way or the other where it's all action and there's no reflection there's no thoughtfulness or it's all talk about talk and there's never really delivery there's never any action so it's really important to me to have a balance there but a balance that's biased towards the verbs i'm on a war against nouns you might actually subtly notice that in my website that i try and get rid of the nouns wherever i can i think the other thing on that is again a very practical concrete step that we recommend in all of the crews that we support to form is to institute a rhythm of reflection intentional reflection so it might be one in five or one in ten of your meetings should be spent looking backwards retrospectively and saying looking back on the last five meetings we had what was good that we wanted to do more what was irritating let's do less of that what are we going to do differently in the next cycle and cultivating that that shared landscape of how do we view this shared project that we're doing and what kind of adjustments can we make and putting that into a rhythmic context where it's a dependable this is another way to reduce uncertainty so we don't know what's going to happen next but we do know that every five meetings we have a retrospective and that that rhythm is dependable and that settles people to make peace with all the change and the uncertainty and that you know the disruption that comes along it's one of the things that you can really count on and rest on so that's like completely essential as a practice for me and i'm always thank my theory i think is always um four or five steps behind my practice i'm always just like let's just do the work and talk about it afterwards well multi-scale nested rhythms happen in the brain like the eeg rhythms as well as a monthly annual etcetera our lab motto is act infer serve because we have to act first that's that critical action orientation and when we act we start to reduce the space of the counterfactuals we're kind of like plucking the strings that are the sinews of the causal structure of the world not that we see the strength themselves but when we observe the outcomes we start to learn more and so when we have to act first though if we start with inference if we start with just theory some people do exit that attractor but also there's so much to think about and we're never going to reduce our uncertainty by thinking alone or we could just reduce our uncertainty and be super precise and incorrect so that's like a very important call if you have any final thoughts please feel free to add them otherwise richard this was an awesome first or extreme and we really appreciate the visit thanks daniel i yeah i could go on many different tangents and rant at great length about plucking the strings of causality which kind of starts to touch into my own homemade cosmology but i think we'll save that we'll save that for later the rant about the cosmic arts

accord maybe i can come back for another conversation sometime very cool so you're always welcome and i hope to continue the conversation with you so again great conversation and good luck with everything you're working on richard thanks so much i would just say to the lovely people listening if you if if you like my way of thinking join me on twitter that's where my stream of consciousness is broadcasting into the void 24 7. um and a more slowed down from from my twitter page richard decibels you'll find like links and other things if you want to go deeper into some other projects and follow websites and things like that great very much recommended so see you later bye